

Q1-Read the questions carefully and choose one correct option:

(30 marks)

1- Using a step size of $h = 0.3$ to solve the differential equation $\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.3) = 5$, then the value $y(0.6)$ using Euler's method is most nearly:

(A) -32.411

(B) 4.577

(C) 3.589

(D) -59.822

2- Applying Newton-Raphson method to estimate the root of $\sqrt{7}$ up to two decimal places. Starting with the initial guess $x_0 = 2$, the first iteration x_1 is:

(A) 1.25

(B) 2.88

(C) 2.75

(D) 2.67

3- The relative error in an approximate solution is 0.004%. The number of significant digits in the solution that we can trust at least is:

(A) 5

(B) 3

(C) 4

(D) 2

4- You are given a system of equations containing a real parameter k :

$x^2 + 12$ and $2x + ky = 22$, then the system has no solution when

(A) $k = 8$

(B) $k = 4$

(C) $k = 2$

(D) $k = 12$

5- The value of the double integral using Trapezoidal rule $\int_2^4 \int_2^4 \frac{x}{y^2} dx dy$ is:
(take the step size $h=1, k=1$)

(A) 0.469

(B) 0.154

(C) 0.108

(D) 0.118

6- Which of the following is a suitable initial interval for computing a positive root of the equation using the bisection method, given that: $g(x) = \cos(x) - 0.5e^x$

(A) $[-1, 0.1]$

(B) $[-1, 0.2]$

(C) $[0.1, 0.4]$

(D) $[0, 0.7]$

7- If the function studied by bisection method changes sign between the initial values 1 and 4 and we know that an error less than 0.00583 is acceptable. Then the required number of iterations that will produce the desired result is:

(A) 8

(B) 12

(C) 14

(D) 9

8- The four-point central difference approximation of the first derivative of the $f(x) = \frac{x^4}{x}$ at $x = 3$ and the step size $h = 1$ is:

(A) 2.767

(B) 0.967

(C) 1.433

(D) 1.800

9- The value of $\int_{8.1}^{8.5} x \cdot \sin(x) dx$ using 1/3 Simpson Rule with $h = 0.2$ is:

(A) 2.973

(B) 9.873

(C) 7.377

(D) 6.110

10- If (0.333) is approximated by a second Lagrange Polynomial for $f(x) = 2^x$ through the points 0, 1, 2, then the approximate value $f(0.333)$ is:

(A) 1.222

(B) 1.275

(C) 1.249

(D) 1.209

1- If A is a 4×4 invertible matrix then what is the inverse of $3A$?

(A) $\frac{1}{4} A^{-1}$

(B) $3 A^{-1}$

(C) $\frac{1}{3} A^{-1}$

(D) $4 A^{-1}$

2- The steepest ascent gradient direction of the function $f(x,y) = y^2 x^2$ at point $(1,2)$ is:

(A) $16i+12j$

(B) $8i+4j$

(C) $12i+16j$

(D) $4i+8j$

3- Consider the following system of linear equation $3x+y-z=12$, $x+5y+2z=14$, $x+4y+9z=10$. The current approximation is $x=4$, $y=2$, $z=0.889$, then the Gauss-Seidal method will give next approximation as $x=3.63$, $y=1.718$ and $z=$

(A) 1.055

(B) 1.014

(C) 0.930

(D) 0.844

4- The value of k_1 by Runge-Kutta method of fourth order if $dy/dx = x + y^2$ and $y(0.2) = 1.1$.

(A) 1.131

(B) 0.566

(C) 0.469

(D) 0.2

5- To find the minimum of the function $f(x) = \sin(x) + x^2$ by golden-section search, assume the first interval is $x_l = -0.8$, $x_u = 0$, then the next interval becomes:

(A) $x_l = 0$, $x_u = -0.4944$

(B) $x_l = -0.8$, $x_u = -0.3056$

(C) $x_l = -0.3056$, $x_u = 0$

(D) $x_l = -0.8$, $x_u = -0.4944$

6- Find the extreme value of y from the following data using Newton's forward interpolation formula.

x	2	3	5
$f(x)$	5	-5	-7

7- Use the second order Romberg integration formula to compute the following integral:

$$\int_2^8 \frac{9\sin(x)}{x} dx$$

⇐ حل امتحان الفيزياء [تقنيات عددية]

⇒ Q1:- المسألة

□ ⇒ $h = 0.3$, $\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.3) = 5$, $y(0.6) = ?$

⇒ sol:- $\frac{dy}{dx} = \sin x - 5y^2$, (حلول الأول لا (Red))

⇒ $y_{i+1} = y_i + f(x_i, y_i)h \rightarrow$ (من الشيفر)

↓
⇒ $y(0.6) = \cancel{5} + f(0.3, 5) * 0.3$

$= 5 + [-124.7 * 0.3]$

$= -32.41 \rightarrow \textcircled{A}$

[من السؤال]

⇒ □ ⇒ $\sqrt{x} = x \Rightarrow x^2 = x \Rightarrow f(x) = x^2 - x$, $x_i = 2$

⇒ $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \rightarrow$ (من الشيفر)

⇒ $f(2) = 4 - 2 = 2$ / $f'(x) = 2x \Rightarrow f'(2) = 4$

⇒ $x_i = 2 - \frac{2}{4} = 1.5 \rightarrow \textcircled{C}$

##

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③ $\Rightarrow \epsilon_a = 0.004\% \Rightarrow \epsilon_s = (0.5 \times 10^{2-n})\% \rightarrow$ (الشك) 6

$\Rightarrow 0.004\% = (0.5 \times 10^{2-n})\% \rightarrow$ (عده الألف)

$\Rightarrow \boxed{n = 4} \rightarrow$ ⑥

④ $\Rightarrow$ solve $x + 2y = 12 / 2x + ky = 22, k = ? \rightarrow$ no. sol...

$\Rightarrow$ no. solution $\xrightarrow{\text{مساواة}}$ $\begin{bmatrix} - & - \\ 0 & 0 \\ \text{zero} & \text{zero} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} - \\ - \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 22 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & k \end{bmatrix} \xrightarrow{-2R_1 + R_2} \begin{bmatrix} 1 & 2 \\ 0 & k-4 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 2 \end{bmatrix} \Rightarrow k = 4 \rightarrow$ ⑤

⑤ $\Rightarrow$ solve $\sum_{i=1}^4 \sum_{j=1}^4 \left(\frac{x}{y^2} \right) \cdot \delta x \delta y \Rightarrow h=1, k=1$

$\Rightarrow$

x/y	2	3	4
2	$\boxed{0.5}$	$\bigcirc \frac{3}{4}$	$\boxed{1}$
3	$\bigcirc \frac{2}{9}$	$\triangle \frac{1}{3}$	$\bigcirc \frac{4}{9}$
4	$\boxed{\frac{1}{8}}$	$\bigcirc \frac{3}{16}$	$\boxed{\frac{1}{4}}$

$\Rightarrow$ $h=2, k=2$ $\leftarrow$ $\frac{x}{y^2}$ من خط الألف وضوء القايير من (2-4)

x/y	2	4
2	$\boxed{0.5}$	$\boxed{1}$
4	$\boxed{\frac{1}{8}}$	$\boxed{\frac{1}{4}}$

$\Rightarrow = \frac{2 \times 2}{4} [1.875 + 0 + 0] = 1.875 \rightarrow$ $\Rightarrow$ $\frac{x}{y^2}$ القانون $\therefore$

$\Rightarrow = \frac{hk}{4} [\text{sum } \square + 2(\text{sum } \bigcirc) + 4(\text{sum } \triangle)]$

$= \frac{1}{4} \left[\frac{15}{8} + [2 * \frac{77}{48}] + [4 * \frac{1}{3}] \right] = \boxed{1.604}$

الجواب منه موجود بالتيارات (السؤال في خط الألف من (1) و (2) و (3) و (4) و (5) و (6) و (7) و (8) و (9) و (10) و (11) و (12) و (13) و (14) و (15) و (16) و (17) و (18) و (19) و (20) و (21) و (22) و (23) و (24) و (25) و (26) و (27) و (28) و (29) و (30) و (31) و (32) و (33) و (34) و (35) و (36) و (37) و (38) و (39) و (40) و (41) و (42) و (43) و (44) و (45) و (46) و (47) و (48) و (49) و (50) و (51) و (52) و (53) و (54) و (55) و (56) و (57) و (58) و (59) و (60) و (61) و (62) و (63) و (64) و (65) و (66) و (67) و (68) و (69) و (70) و (71) و (72) و (73) و (74) و (75) و (76) و (77) و (78) و (79) و (80) و (81) و (82) و (83) و (84) و (85) و (86) و (87) و (88) و (89) و (90) و (91) و (92) و (93) و (94) و (95) و (96) و (97) و (98) و (99) و (100)

6 $\Rightarrow g(x) = \cos(x) - 0.5 e^x$ / لازم رقمی می باشد
و لا یسر (-) باشد

$\Rightarrow [0, 0.7] \Rightarrow g(0) = \cos(0) - 0.5 e^0 \Rightarrow g(0) = 0.5 \rightarrow (+)$
 $\Rightarrow g(0.7) = \cos(0.7) - 0.5 e^{0.7} \Rightarrow g(0.7) = -0.242 \rightarrow (-)$
 $\downarrow$
 $\checkmark$ ①

$\Rightarrow$ ② $\Rightarrow$ solve $[2, 4]$, $\epsilon_e = 0.00583$

$\Rightarrow n \geq \frac{\ln(b-a) - \ln(\epsilon_e)}{\ln(2)} \rightarrow$ (مدت التكرار)

$\Rightarrow n \geq \frac{\ln(4-2) - \ln(0.00583)}{\ln(2)} \Rightarrow n \geq 8.4223$

$\Rightarrow n \geq 9 \rightarrow$ ①

ما را در حالت
تکرار من (8)
منقذ بها لأقدم عدد
الحاصل

$\Rightarrow$ ③ $\Rightarrow$ solve ④ $\rightarrow$ point (نقطه) $\rightarrow$ center . d . A

$f(x) = \frac{2^x}{x}$, $x_i = 3$, $h = 1$.

$\Leftarrow$ به ازای نقطه و قانون (center) $\Leftarrow$ مستخدم الا نه من التالي : (من الشيفرة)

$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h} \rightarrow$ [التي اشتغلها
لأنه بدو
نقطة]

$\Rightarrow x_i = 3$, $x_{i+1} = 4$, $x_{i+2} = 5$, $x_{i-1} = 2$, $x_{i-2} = 1 \rightarrow$
 $\Rightarrow f'(x_i) = \frac{-f(5) + 8f(4) - 8f(2) + f(1)}{12} = 0.967 \rightarrow$ ②

احتمالاً بدو نقطه
بمقدار (h)

$$\textcircled{9} \Rightarrow \text{sol'n } \int_{8.1}^{8.5} \sqrt{x \sin(x)} \cdot dx' \Rightarrow h = 0.2$$

$$[x_0 = 8.1, x_1 = 8.3, x_2 = 8.5]$$

$$\Rightarrow h = \frac{8.5 - 8.1}{0.2} = 2 \rightarrow \text{Simpson Rule (simple)}$$

$$\Rightarrow \int_{x_0}^{x_2} f(x) \cdot dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$$\approx \frac{0.2}{3} [f(8.1) + [4 * f(8.3)] + f(8.5)]$$

$$\approx \frac{0.2}{3} [7.8561 + [4 * 7.488] + 6.78714]$$

$$\approx 2.973 \rightarrow \textcircled{A}$$

$$\textcircled{10} \Rightarrow \text{sol'n } f(x) = 2^x, \text{ points } \rightarrow 0, 1, 2$$

نقاط 3 نقاط $\therefore$ الإقتراح لدرجة 2 (من الدرجة 2)
 عدد النقاط - 1 = درجة الإقتراح

$$\Rightarrow f(0.333) = \frac{(0.333-1)(0.333-2)}{(0-1)(0-2)} f(0) + \frac{(0.333-0)(0.333-2)}{(1-0)(1-2)} f(1)$$

$$+ \frac{(0.333-0)(0.333-1)}{(2-0)(2-1)} f(2) = 1.222 \rightarrow \textcircled{A}$$

$$\textcircled{11} \Rightarrow [3A]^{-1} \xrightarrow{\text{inverse}} \frac{1}{3} A^{-1}$$

$$\textcircled{12} \Rightarrow f(x, y) = y^2 x^2, (1, 2)$$

$$\Rightarrow \frac{\partial f}{\partial x} = 2xy^2, \frac{\partial f}{\partial y} = 2yx^2$$

$$\Rightarrow (1, 2) \Rightarrow \nabla f = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \Rightarrow = 8\hat{i} + 4\hat{j} \quad \textcircled{B}$$

⇒ 13 ⇒ solve $x + 4y + 9z = 20$

⇒ $z = \frac{20 - x - 4y}{9}$, $x = 3.63$, $y = 1.718$

⇒ $z = \frac{20 - 3.63 - (4 * 1.718)}{9} = 1.055 \rightarrow \textcircled{A}$

⇒ 14 ⇒ solve $\frac{dy}{dx} = x + y^2$, $y(0.2) = 2.34$

⇒ $h = 0.1$ / ⇒ $k_1 \rightarrow$ Runge-Kutta $\rightarrow 4^{th}$ order

⇒ $k_1 = h * f(x_i, y_i) \rightarrow$ (مشتق)

مشتق
من قيمه y
(1)

⇒ $k_1 = 0.1 * f(0.2, 2.34)$

$= 0.1 * 5.67 = 0.567 \approx 0.566 \rightarrow \textcircled{B}$

⇒ 15 ⇒ solve ⇒ minimum / $f(x) = \sin(x) + x^2$ / $[x_L, x_u] = [-0.8, 0]$

⇒ $d = 0.618033 (x_u - x_L) = 0.618033 * (0 - (-0.8)) = 0.4944264$

⇒ $x_1 = x_L + d = -0.8 + 0.4944264 = -0.3055736$

⇒ $x_2 = x_u - d = 0 - 0.4944264 = -0.4944264$

⇒ $f(x_1) = -0.207465$ / $f(x_2) = -0.230069$

⇒ minimum ⇒ $f(x_1) > f(x_2) \Rightarrow x_1 = x_u$

⇒ $[x_L = -0.8 / x_u = x_1 = -0.3056] \Rightarrow \textcircled{B}$

⇒ Q.2 ⇒ solve extreme value ~~max/min~~

⇒

x	f(x)	b ₁	b ₂
2	5	-10	3
3	-5	-1	
5	-7		

$$\Rightarrow b_1 = f(3, 2) = \frac{f(3) - f(2)}{3 - 2} = \frac{-5 - 5}{1} = -10$$

$$\Rightarrow b'_1 = f(5, 3) = \frac{f(5) - f(3)}{5 - 3} = \frac{-7 + 5}{2} = \frac{-2}{2} = -1$$

$$\Rightarrow b_2 = f(5, 3, 2) = \frac{-1 + 10}{5 - 2} = \frac{9}{3} = 3$$

$$\Rightarrow f(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$\Rightarrow f(x) = 5 + 10(x - 2) + 3(x - 2)(x - 3)$$

$$= 5 - 10x + 20 + 3(x^2 - 3x - 2x + 6)$$

$$= 5 - 10x + 20 + 3x^2 - 15x + 18$$

$$f(x) = 3x^2 - 25x + 43$$

$$f'(x) = 6x - 25 \Rightarrow 6x - 25 = 0 \Rightarrow x = 4.167 \quad \#$$

$$\Rightarrow f''(x) = 6 \Rightarrow f''(4.167) = 6 \quad \#$$

2.3 i

$$\Rightarrow \int_2^8 \frac{9 \sin(x)}{x} \cdot dx \quad / \Rightarrow n=1 \Rightarrow \text{second order}$$

• clear data $\leftarrow (4, 2)$

Trapezoidal

$$\Rightarrow n=1 \Rightarrow h = \frac{b-a}{n} = \frac{8-2}{1} = 6 \Rightarrow \hat{I} = \frac{6}{2} [f(2) + f(8)]$$

$$\Rightarrow R(1,0) = 3 [4.09184 + 1.11303] = 15.61461$$

$$\Rightarrow n=2 \Rightarrow h = \frac{8-2}{2} = 3 \Rightarrow R(2,0) = \frac{3}{2} [f(2) + 2f(5) + f(8)]$$

$$\Rightarrow R(2,0) = \frac{3}{2} [4.09184 + [2 \times -1.72606] + 1.11303] = 2.62913$$

$$\Rightarrow n=4 \Rightarrow h = \frac{8-2}{4} = \frac{6}{4} \Rightarrow R(3,0) = \frac{6}{2} [f(2) + 2f(3.5) + f(5) + 4f(6)]$$

$$+ f(8)] = \frac{6}{2} [4.09184 + [2 \times (-0.90201) + -1.72606 + 0.29786]$$

$$+ 1.11303] = \cancel{0.40834} \quad 0.40834$$

$$\Rightarrow n=6 \Rightarrow h = \frac{8-2}{6} = 1 \Rightarrow R(4,0) = \frac{1}{2} [f(2) + [2(f(3) + f(4) + f(5) + f(6) + f(7))$$

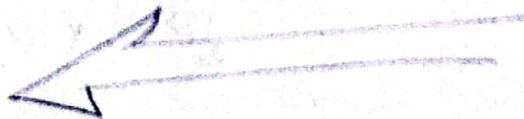
$$+ f(8)]$$

$$\Rightarrow R(4,0) = \frac{1}{2} [4.09184 + [2(0.42336 + -1.70281 + -1.72606 + 0.41912$$

$$+ 0.84470) + 1.11303] = 0.045755 \quad \times$$

{clear data}

$$[R(i,k) = \frac{4^k R(i,k-1) - R(i,k-1)}{4^k - 1}]$$



$R(i, k)$	First order	Second order
$[15.61461]$ $\hat{R}(1,0)$		
$[2.62913]$ $\hat{R}(2,0)$	-1.699363333	
$[0.40834]$ $\hat{R}(3,0)$	-0.3319233333	-0.2407573333
$[0.04575]$ $\hat{R}(4,0)$	-0.07510666667	-0.05798933333

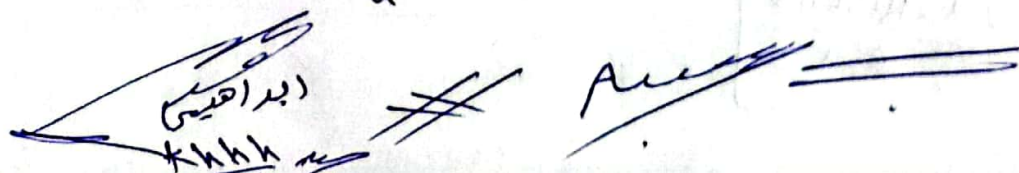
$$\Rightarrow R(2,1) = \frac{[4^{(1)} * R(2,0) - R(1,0)]}{4^{(1)} - 1} = -1.69936$$

$$\Rightarrow R(3,1) = \frac{[4^{(1)} * R(3,0) - R(2,0)]}{4^{(1)} - 1} = -0.33192$$

$$\Rightarrow R(4,1) = \frac{[4^{(1)} * R(4,0) - R(3,0)]}{4^{(1)} - 1} = -0.07511$$

$$\Rightarrow R(3,2) = \frac{[4^{(2)} * R(3,1) - R(2,1)]}{4^{(2)} - 1} = -0.24076$$

$$\Rightarrow R(4,2) = \frac{[4^{(2)} * R(4,1) - R(3,1)]}{4^{(2)} - 1} = -0.05799$$



Q1-Answer the following questions (18marks)

The determinant of hessian of the function $x^2 - 2y^2 - 4y + 6$ at point $(0, 0)$ is

Answer:

Given $f(4) = 3$, $f'(4) = 8$, $f''(4) = 11$, and that all other higher order derivatives of $f(x)$ are zero at $x = 4$, and assuming the function and all its derivatives exist and are continuous between $x = 4$ and $x = 5$, the value of $f(5)$ is

Answer:

The root of the equation $f(x) = 0$ is found by using the Newton-Raphson method. The initial estimate of the root is $x_0 = 3$, $f(3) = 5$. The angle the line tangent to the function $f(x)$ makes at $x = 3$ is 57° with respect to the x-axis. The next estimate of the root, x_1 most nearly is

Answer:

Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -9 & -11.3 \\ 10.3 & -7 & 8.5 \end{bmatrix}, [B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 12 & -20 \end{bmatrix} \text{ then if } [C] = [A][B], \text{ then } C_{31} =$$

Answer:

The lower triangular matrix $[L]$ in the $[L][U]$ decomposition of matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \text{ is}$$

Answer:

The following x, y data is given.

x	6	9	14
y	24	37	25

The Newton's divided difference second order polynomial for the data is given by $f_2(x) = b_0 + b_1(x - 6) + b_2(x - 6)(x - 9)$. The value of b_1 is most nearly

Answer:

The following

x	1	2	3	4
y	1	200	400	900

data is regressed with least squares regression to $y = a_0 + a_1x$. The value of a_1 most nearly is

Answer:

The value of $\int_{0.4}^{2.0} e^x dx$ by using 4-segment Simpson's $1/3$ rule most nearly is

Answer:

The value of an integral $\int_a^b f(x)dx$ given using 1, 2, and 4 segments trapezoidal rule is given as 6.3460, 2.4408, and 1.5536, respectively. The best estimate of the integral you can find using Romberg integration is most nearly

Answer:

Given $4\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.2) = 9$
and using a step size of $h = 0.3$, the value of $y(0.8)$ using Euler's method is most nearly

Answer:

Given $2\frac{dy}{dx} + y^2 = e^x$, $y(0.6) = 8$, and using a step size of $h = 0.3$, the best estimate of $\frac{dy}{dx}(0.9)$ Runge-Kutta 4th order method is most nearly:

Answer:

Q2-Determine the minimum of the function $f(x, y) = x^2 + y^2$. Use point (2, 1) as the initial estimate of the optimal solution. Conduct one iteration. (12marks)

Q3-Evaluate

$$I = \int_2^{2.6} \int_4^{4.9} \frac{1}{xy} dx dy$$

Take $dx = 0.2$ in x-direction, $dy = 0.3$ in y-direction. Use Trapezoidal rule in x - direction and Simpson's 3/8 rule in y-direction.

(10marks)

Q4-Determine the values of $\underline{a}$ and $\underline{b}$ for which the system (10marks)

$$\begin{bmatrix} 3 & 0 & 0 \\ 9 & 3 & 2 \\ 1 & 1 & a \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$$

1) has a unique solution 2) has no solution and 3) has infinitely many solutions

{Q.2} $\Rightarrow$ Soln Taylor series ~~of~~ $f(x)$

$$\Rightarrow f(x_{i+1}) = f(x_i) + \frac{f'(x_i)}{1!} h + \frac{f''(x_i)}{2!} h^2$$

$(5-4=1)$

$$\Rightarrow f(5) = f(4) + \frac{f'(4)(1)}{1} + \frac{f''(4)(1)^2}{2}$$

$$= 3 + 8 + \frac{11}{2}$$

$$f(5) = \underline{\underline{16.5}} \quad \times$$

مسألة (المسألة الأولى)

حل: $x^2 - 2y^2 - 4y + 6$

$$\Rightarrow \frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = -4y - 4$$

$$\nabla f = \begin{bmatrix} 2x(0,0) \\ -4y - 4(0,0) \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \end{bmatrix}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = -4, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 0$$

$$\Rightarrow H = \begin{bmatrix} 2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$\begin{aligned} |H| &= (2 \times -4) - (0 \times 0) \\ &= -8 \end{aligned}$$

⇒

SMURFS POLYTECHNIC UNION

③ ⇒ solⁿ ⇒ root → $f(x) = 0$

⇒ $x_0 = 3$ / $f(3) = 5$

$f(3)$
^

⇒ $x = 3 \rightarrow 52^\circ \rightarrow \tan(52^\circ) = \frac{f(3)}{f'(3)} = \frac{5}{0.484699}$

⇒ $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$

$= 3 - \frac{f(3)}{f'(3)} = 3 - \frac{5}{0.484699}$

$= -7.315680$

④

$$[C] = [A] [B]$$

↙ $2 \times 3 \times 2 \times 3$

$$\Rightarrow \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -9 & 11.3 \\ 10.3 & -7 & 8.5 \end{bmatrix} \cdot \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 12 & -20 \end{bmatrix} = \begin{bmatrix} 329.7 & -430.6 \\ -68 & 217.2 \\ 157.6 & -170.8 \end{bmatrix}$$

$3 \times 3 \quad 2 \times 2 \quad 3 \times 2$

$$C_{31} = 157.6$$

⑤

$$[A] = [L] [U]$$

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix}$$

$$\Rightarrow L_{21} = \frac{a_{21}}{a_{11}} = \frac{10}{25} = 0.4$$

$$L_{31} = \frac{a_{31}}{a_{11}} = \frac{8}{25} = 0.32$$

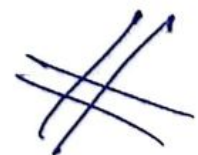
$$\Rightarrow \begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix} \Rightarrow -2R_1 + 5R_2$$

$$\begin{bmatrix} 25 & 5 & 4 \\ 0 & 30 & 52 \\ 8 & 12 & 22 \end{bmatrix} \Rightarrow -8R_1 + 25R_3$$

$$\begin{bmatrix} 25 & 5 & 4 \\ 0 & 30 & 52 \\ 0 & 260 & 518 \end{bmatrix} \rightarrow \begin{bmatrix} a_{11}' & a_{21}' & a_{31}' \\ a_{12}' & a_{22}' & a_{32}' \\ a_{13}' & a_{23}' & a_{33}' \end{bmatrix}$$

$$L_{32} = \frac{a_{32}}{a_{22}} = \frac{260}{30} = 8.66667$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0.4 & 1 & 0 \\ 0.32 & 8.6667 & 1 \end{bmatrix}$$



⇒ ⑥

⇒ sol:

المسألة

x	y	b ₁	b
6	24		
9	37		
14	25		

$$\begin{aligned}
 \Rightarrow b_1 &= \frac{\sum(x_i y_i) - \sum(x_i) \sum(y_i)}{\sum(x_i^2) - (\sum(x_i))^2 / n} \\
 &= \frac{37 - 24}{9 - 6} = 4.3333
 \end{aligned}$$

⑦

⇒ sol: $y = a_0 + a_1 x$

$$\begin{bmatrix} \sum x_i & \sum x_i^2 \\ \sum x_i y_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

المسألة

$$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 150 \\ 520 \end{bmatrix}$$

$$\Rightarrow 4a_0 + 10a_1 = 150$$

$$10a_0 + 30a_1 = 520$$

$$\Rightarrow \textcircled{8} \int_{0.4}^{2.0} e^x \cdot dx \quad / \quad n=4 \quad / \quad \textcircled{1/3}$$

(نقطة رقيقة ونا حكيمة)

(5) نقطة

$$\Rightarrow 5.01 \dots \quad n=4$$

$$x_0 = 0.4$$

$$x_4 = 2$$

$$h = \frac{b-a}{n}$$

$$= \frac{2-0.4}{4}$$

$$= 0.4$$

$$x_0 = 0.4$$

$$x_1 = 0.8$$

$$x_2 = 1.2$$

$$x_3 = 1.6$$

$$x_4 = 2$$

$$\int_a^b f(x) \cdot dx = \frac{h}{3} \left[f(x_0) + 2(f(x_1) + f(x_2) + f(x_3) + f(x_4)) \right]$$

$$+ 4(f(x_1) + f(x_3)) + f(x_4)$$

$$= \frac{0.4}{3} \left[1.491825 + 2(3.320149) \right]$$

$$+ 4(2.25541 + 4.95303) + 7.38906$$

$$5.898052$$

$\Rightarrow \textcircled{9} \Rightarrow \text{sols } n = 1, n = 2, n = 3$

$\Rightarrow$ (درجہ اولیٰ) $\leftarrow$ (درجہ اولیٰ) $n = 3$ $\leftarrow$

$\Rightarrow$ ~~Table~~

(i, k)	1 st	2 nd
$R(1, 0)$		
$R(2, 0)$	$R(2, 1)$	Table
$R(3, 0)$	$R(3, 1)$	$R(3, 2)$

111

$\leftarrow$ ~~Table~~

(i, k)	1 st	2 nd
6.3460		
2.4408	2.736533	
1.5536		

$$\begin{aligned}
 R(2, 1) &= \frac{4 \times R(2, 0) - R(1, 0)}{4 - 1} \\
 &= \frac{4 \times 2.4408 - 6.3460}{3} \\
 &= \del{2.736533} \\
 &= 1.138066
 \end{aligned}$$

$$\Rightarrow R(3,1) = \frac{4^1 \cdot R(3,0) - R(2,0)}{4^1 - 1}$$

$$= \frac{4^1 \cdot 1.5536 - 2.4408}{3}$$

$$= 1.252862$$

$$\Rightarrow R(3,2) = \frac{4^2 \cdot R(3,1) - R(2,1)}{4^2 - 1}$$

$$= \frac{16 \cdot 1.252862 - 1.139068}{16 - 1}$$

$$= 1.265282$$

(16) $\Rightarrow$ solⁿ $4 \frac{dy}{dx} + 5y^2 = \sin x$

$\Rightarrow \frac{dy}{dx} = \frac{\sin x - 5y^2}{4}$ / $y = 0$
 $x = 0.2$
 $h = 0.3$

Euler $\rightarrow y_{i+n} = y_i + f(x_i, y_i)h$
 $\Rightarrow$ (2nd order) initial value

$\Rightarrow \square 1 \Rightarrow y(0.5) = y(0.2) + f(0.2, y)h$
 $= 0 + (-10.2 \times 0.3)$
 $= -21.36$

$\Rightarrow \square 2 \Rightarrow y(0.8) = y(0.5) + f(0.5, -21.36)h$
 $= -21.36 + (-570.19 \times 0.3)$
 $\therefore -192.417$

$\Rightarrow$ (11) $\Rightarrow$ solve RK(4)

$$\Rightarrow 2 \frac{dy}{dx} + y^2 = e^x$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^x - y^2}{2} \quad \begin{array}{l} y = 8 \\ x = 0.6 \\ h = 0.3 \end{array}$$

2 step

$$\Rightarrow k_1 = h f(x_i, y_i) = 0.3 f(0.6, 8) \\ = 0.3 \times 26682 \\ = \text{~~7.9046~~}$$

$$\Rightarrow k_2 = h f(x_i + 0.5h, y_i + 0.5k_1) \\ = 0.3 f(0.75, \text{~~7.9046~~}) \\ = \text{~~2.5146~~}$$

- 5.08146 h

$$k_3 = h f(x_i + 0.5h, y_i + 0.5k_2)$$

$$= 0.3 f(0.75, 5.245926)$$

$$= -15.44933681$$

$$\Rightarrow k_4 = h f(x_i + h, y_i + k_3)$$

$$= 0.3 f(0.9, -2.24933981)$$

$$= -26.51653025$$

$$\Rightarrow y(0.9) = 8 \times \frac{1}{6} \left\{ \begin{aligned} &-9.326682 \\ &+ 2(-4.5081466) \\ &+ 2(-15.44933681) \\ &+ -26.5163025 \end{aligned} \right\}$$

$$= -4.62632522$$

Answer the following questions

Q1- Fit a straight-line to the data provided in the following Table:

x_i	5	7	9	11	13
y_i	6	4.5	3.4	2.3	1.4

(10 marks)

The equation of the line is:

The value of coefficient of determination, r^2 is:

Q2- Compute the integration: $\int_2^4 \frac{1}{(x-1)^2} dx$, using Work to five significant figures (10 marks)

Apply Romberg integration taking $n=1$, $n=2$ & $n=4$ up to $R(3, 2)$.

Answer:

.....

Q3- Evaluate the following double integration: $\int_1^3 \int_1^3 (2x + y) dx dy$. (10 marks)

Take $dx = 0.5$ in the x -direction, $dy = 0.5$ in the y -direction. Apply the Trapezoidal rule in the x -direction and in the y -direction.

Answer:

.....

Q4- Maximize the function: $f(x, y) = xy + 4x - x^2 - y^2 - 4$ (10 marks)

Apply the Optimal steepest ascent method, use point (2, 2) as the initial estimate of the optimal solution. The optimum point after conducting one iteration is:

Answer:

.....

Q5- Consider the ordinary differential equation: $\frac{dy}{dx} = x^2 - 2y^2 + x$ (10 marks)

Given: $x_0 = 1$, $y_0 = 2$, $h = 0.5$

Use the Runge-Kutta method to the 4th order to find $y(1.5)$.

Answer:

.....

==> المعادلات القياسية <==> تقريب عددي

Q.1 $\Rightarrow$ solve $y = a_0 + a_1 x$ $\Rightarrow y = 8.51 - 0.55x$

$\Rightarrow a_0 = \bar{y} - a_1 \bar{x}$

$a_0 = 3.56 - 9a_1 \Rightarrow a_0 = 3.56 - (9 \times -0.55)$
 $\Rightarrow a_0 = 8.51$

$\bar{y} = \frac{\sum y}{n} = 3.56$

$\bar{x} = \frac{\sum x}{n} = 9$

$\Rightarrow a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} = \frac{(5 \times 138.2) - (45 \times 17.8)}{(5 \times 445) - (2025)} = -0.55$

x_i	y_i	$x_i y_i$	$(y - \bar{y})^2 \rightarrow "S_t"$	$(y_i - a_0 - a_1 x_i)^2 \rightarrow "S_r"$
5	6	30	5.9536	0.0576
7	4.5	31.5	0.8836	0.0256
9	3.4	30.6	0.0256	0.0256
11	2.3	25.3	1.6876	0.0256
13	1.6	20.8	3.8416	0.0576
Σ	45	138.2	12.292	

$\Rightarrow r^2 = \frac{S_t - S_r}{S_t} = \frac{\sum (y - \bar{y})^2 - S_r}{\sum (y - \bar{y})^2} = \frac{12.292 - 0.192}{12.292}$

~~12.292~~ = 0.98438

= 98.44%

Q.2 $\Rightarrow$ حل :- $\int_2^4 \frac{1}{(x-1)^2} dx$

$\Rightarrow$ $\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i) \Delta x$ $\leftarrow$ (ن) قسم $\Delta x =$

$\Rightarrow n=1 \Rightarrow h = \frac{b-a}{n} = \frac{4-2}{1} = 2 \quad \left/ \begin{array}{l} x_0=2 \\ x_1=4 \end{array} \right.$

$\Rightarrow R(1,0) = \frac{h}{2} [f(x_0) + f(x_1)] = \frac{2}{2} [f(2) + f(4)]$
 $= 1 [1 + 1.11111] = 2.11111$

$\Rightarrow n=2 \Rightarrow h = \frac{b-a}{n} = \frac{4-2}{2} = 1 \quad \left/ \begin{array}{l} x_0=2 \\ x_1=3 \\ x_2=4 \end{array} \right.$

$R(2,0) = \frac{h}{2} [f(x_0) + 2f(x_1) + f(x_2)] = \frac{1}{2} [f(2) + 2f(3) + f(4)]$
 $= 0.5 [1 + 2 \times 1.11111 + 1.11111] = \cancel{1.30556} 1.30556$

$\Rightarrow n=4 \Rightarrow h = \frac{b-a}{n} = \frac{4-2}{4} = 0.5 \quad \left/ \begin{array}{l} x_0=2 \\ x_1=2.5 \\ x_2=3 \\ x_3=3.5 \\ x_4=4 \end{array} \right.$

$\Rightarrow R(4,0) = \frac{h}{2} [f(x_0) + 2[f(x_1) + f(x_2) + f(x_3) + f(x_4)]]$
 $= \frac{0.5}{2} [f(2) + 2[f(2.5) + f(3) + f(3.5)] + f(4)]$
 $= \frac{0.5}{2} [1 + 2[0.44444 + 0.25 + 0.16] + 1.11111]$
 $= \cancel{0.95499} 0.954997$

$$\Rightarrow R(2,1) = \frac{[4^1 \times R(2,0) - R(1,0)]}{4^1 - 1}$$

$$= \frac{[4 \times 1.30556 - 2.1111]}{3} = 1.03704$$

$$\Rightarrow R(3,1) = \frac{[4^1 \times R(3,0) - R(2,0)]}{4^1 - 1}$$

$$= \frac{[4 \times 0.954997 - 1.30556]}{3} = 0.83814$$

$$\Rightarrow R(3,2) = \frac{[4^2 \times R(3,1) - R(2,1)]}{4^2 - 1}$$

$$= \frac{[16 \times 0.83814 - 1.03704]}{15} = 0.82488$$

$\Rightarrow$

(i, k)	1 st	2 nd
2.11111		
1.30556	1.03704	
0.954997	0.83814	0.82488

#

$$\Rightarrow \underline{Q.3} \Rightarrow \text{sol'n} \quad \int_1^3 \int_1^3 (2x+y) dx dy$$

$$\Rightarrow \Delta x = h = 0.5 \Rightarrow x_0 = 1, x_1 = 1.5, x_2 = 2, x_3 = 2.5, x_4 = 3$$

$$\Rightarrow \Delta y = k = 0.5 \Rightarrow y_0 = 1, y_1 = 1.5, y_2 = 2, y_3 = 2.5, y_4 = 3$$

$\Rightarrow$

$y \backslash x$	1	1.5	2	2.5	3	—
1	3	4	5	6	7	—
1.5	3.5	4.5	5.5	6.5	7.5	—
2	4	5	6	7	8	—
2.5	4.5	5.5	6.5	7.5	8.5	—
3	5	6	7	8	9	—

$$\Rightarrow \text{in } x\text{-direction (Trapezoidal Rule)} \Rightarrow I = \frac{h}{2} [f(x_0) + 2\{f(x_1) + f(x_2) + f(x_3)\} + f(x_4)]$$

$$\Rightarrow y = 1 \Rightarrow I = \frac{0.5}{2} [3 + 2\{4 + 5 + 6\} + 7] = 10$$

$$\Rightarrow y = \cancel{1.5} \Rightarrow I = \frac{0.5}{2} [3.5 + 2\{4.5 + 5.5 + 6.5\} + 7.5] = 11$$

$$\Rightarrow y = 2 \Rightarrow I = \frac{0.5}{2} [4 + 2\{5 + 6 + 7\} + 8] = 12$$

$$\Rightarrow y = 2.5 \Rightarrow I = \frac{0.5}{2} [4.5 + 2\{5.5 + 6.5 + 7.5\} + 8.5] = 13$$

$$\Rightarrow y = 3 \Rightarrow I = \frac{0.5}{2} [5 + 2\{6 + 7 + 8\} + 9] = 14$$

$$\Rightarrow \text{in } y\text{-direction (Trapezoidal Rule)}$$

$$\Rightarrow I = \frac{h}{2} [f(y_0) + 2\{f(y_1) + f(y_2) + f(y_3)\} + f(y_4)]$$

$$= \frac{0.5}{2} [10 + 2\{11 + 12 + 13\} + 14]$$

$$= 24$$

Q. 4

$$\Rightarrow \text{sol}^n \quad f(x, y) = xy + 4x - x^2 - y^2 - 4$$

$$\Rightarrow \frac{\partial f}{\partial x} = y + 4 - 2x \quad / \quad \frac{\partial f}{\partial y} = x - 2y$$

$$\Rightarrow \nabla f = \begin{bmatrix} y + 4 - 2x \\ x - 2y \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$\downarrow$
(2, 2)

$$\Rightarrow x' = x_i + \nabla f_x h = 2 + 2h$$

$$\Rightarrow y' = y_i + \nabla f_y h = 2 - 2h$$

$$\Rightarrow g(h) = (2 + 2h)(2 - 2h) + 4(2 + 2h) - (2 + 2h)^2 - (2 - 2h)^2 - 4$$

$$= (4 - 4h + 4h - 4h^2) + (8 + 8h) - (4 + 8h + 4h^2) - (4 - 8h + 4h^2) - 4$$

$$= 4 - 4h^2 + 8 + 8h - 4 - 8h - 4h^2 - 4 + 8h + 4h^2 - 4$$

$$g(h) = -4h^2 + 8h \Rightarrow g'(h) = -8h + 8$$

$$\Rightarrow -8h + 8 = 0 \Rightarrow \frac{8h}{8} = \frac{8}{8} \Rightarrow \boxed{h = 1}$$

$$\Rightarrow x' = 2 + (2 \times 1) = 4 \quad / \quad y' = 2 - (2 \times 1) = 0$$

$$\Rightarrow \text{next point} \rightarrow (4, 0)$$

$$\Rightarrow \underline{Q.5} \Rightarrow \text{sol: } \frac{dy}{dx} = x^2 - 2y^2 + x$$

$$\Rightarrow x_0 = 1, y_0 = 2, h = 0.5 \xrightarrow{\text{step 1}} \underline{\text{step 2}}$$

$$\Rightarrow k_1 = h f(x_0, y_0) = 0.5 f(1, 2) = -3$$

$$\begin{aligned} \Rightarrow k_2 &= h f(x_0 + 0.5h, y_0 + 0.5k_1) = 0.5 f(1 + (0.5 \times 0.5), 2 + (0.5 \times -3)) \\ &= 0.5 f(1.25, 0.5) = 1.15625 \end{aligned}$$

$$\begin{aligned} \Rightarrow k_3 &= h f(x_0 + 0.5h, y_0 + 0.5k_2) = 0.5 f(1 + (0.5 \times 0.5), 2 + (0.5 \times 1.15625)) \\ &= 0.5 f(1.25, 2.578125) = -5.24045 \end{aligned}$$

$$\begin{aligned} \Rightarrow k_4 &= h f(x_0 + h, y_0 + k_3) = 0.5 f(1 + 0.5, 2 + (-5.24045)) \\ &= 0.5 f(1.5, -3.24045) = 5.11545 \end{aligned}$$

$$\begin{aligned} \Rightarrow y(1.5) &= y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ &= 2 + \frac{1}{6} (-3 + 2 \times 1.15625 - 5.24045) + 5.11545 \end{aligned}$$

$$\Rightarrow y(1.5) = 0.991175$$

Step 2